GEOSPATIAL RAP TOOL

Developer and Administrator Guide

Spatial configuration, architecture, deployment, and maintenance

Documentation revision	June 2026
Language	English
Status	Aligned with the supplied current application source

Purpose. This document describes the current workflow and implementation represented by the supplied project files. It replaces older instructions that referred to previous navigation, storage, or output structures.

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1. Current project structure

```
Geospatial-RAP/
|-- app.R
|-- global.R
|-- config.json
|-- Geospatial-RAP.Rproj
|-- raw_data/                # Deploy configured .gpkg files here
|-- R/
|   |-- help_ui.R
|   |-- shared/{api_pop_io.R, config_helpers.R}
|   |-- geospatial/
|   |   |-- functions_io.R
|   |   |-- functions_spatial.R
|   |   |-- functions_plot.R
|   |   |-- mod_geospatial.R
|   |   `-- mod_geospatial/{uploads,inputs,maps,tables,downloads,helpers}.R
|   `-- scripts/main.R
|-- tests/
`-- www/{config.css, docs/}
```

global.R disables s2 geometry for planar compatibility, loads configuration, resolves active countries and defaults, performs geospatial pre-flight checks, and then sources the remaining modules.

Component	Responsibility
functions_io.R	Input standardisation and geospatial aggregation.
functions_spatial.R	GeoPackage loading, spatial join, neighbours, Global Moran and LISA.
functions_plot.R	Leaflet prevalence and LISA maps, legends, popups, and result-table preparation.
mod_geospatial/inputs.R	Country, geography, time controls, validation, preparation, layer load, and join.
uploads.R	Uploaded-file manager.
maps.R	Moran banner and leaflet outputs.
tables.R	Filterable results/prepared tables and preparation log.
downloads.R	CSV and optional map exports.
config_helpers.R	Country defaults, mapping helpers, GeoPackage paths, and pre-flight checks.

2. Local execution and dependencies

1. Place the configured GeoPackage files in raw_data/.
2. Open Geospatial-RAP.Rproj.
3. Install or restore packages.
4. Run app.R or shiny::runApp() from the project root.

Area	Packages
Shiny/UI	shiny, shinyjs, DT, htmltools, jsonlite
Data	dplyr, tidyr, purrr, rlang, readr, stringr, lubridate
Spatial	sf, spdep, leaflet

Geometry mode. The supplied global.R calls sf::sf_use_s2(FALSE). Test geometry validity and neighbour results when changing spatial libraries or upgrading sf/spdep.

3. Geospatial configuration

Key	Purpose
is_offline	Controls map behavior for offline deployments.
active_country / default_country	Default country selection.
default_level	Preferred geographic level when available.
default_year	Optional default epidemiological year.
target_var	Name of the calculated measure, currently prevalence_100k.
geospatial.rate_multiplier	Multiplier used in the prevalence calculation, currently 100000.
Country.file_name	GeoPackage filename expected under raw_data/.
Country.levels	Ordered selectable geographic levels.
mapping.<level>.csv_column	Generic CSV column used for that level.
layer_name	GeoPackage layer.
spatial_id	Unique polygon identifier.
name_col / label_col	Human-readable polygon label.
code_length	Optional fixed identifier width for zero padding.
ui_label	Optional user-facing level label.

```
  "department": {
    "csv_column": "level2",
    "layer_name": "Department",
    "spatial_id": "cde",
    "name_col": "nam",
    "code_length": 5
  }
```

rap_geospatial_preflight() checks the configured file, layer names, spatial ID columns, and name/label columns. Errors are displayed in the UI and block analysis until corrected.

Storage change. The current application reads GeoPackage layers directly from raw_data/<file_name>.gpkg. The former optimized .rds/build_data workflow no longer describes this implementation.

4. Data pipeline

5. api_pop_combine_files() reads and standardises active CSV files with require_population = TRUE.
6. prepare_geospatial_dataset() filters one year and inclusive week window.
7. Cases are summed by the selected analysis unit.
8. Population is deduplicated at the most detailed available geography before roll-up.
9. The target measure is calculated as n_cases / population * rate_multiplier.
10. load_geospatial_layer() reads only the configured ID, label, and geometry, makes geometry valid, and transforms to EPSG:4326.
11. join_analysis_to_geometry() matches prepared IDs to map IDs or labels.

The prepared dataset retains a log describing filtering, aggregation, denominator handling, rate calculation, and unit count.

Population semantics. The implementation assumes population is a static denominator repeated across weeks. If population varies within the selected period, the deduplication rule must be reviewed before release.

5. Spatial analysis implementation

get_spatial_weights() calls spdep::poly2nb() using its default polygon-contiguity behavior and converts the neighbour list to row-standardised W weights with zero.policy = TRUE.

compute_global_moran() returns NULL when fewer than four units are available, all values are missing, variance is zero, or the test fails. Otherwise it uses spdep::moran.test().

compute_lisa() uses spdep::localmoran(), standardises the target variable, calculates its spatial lag, and assigns High-High, Low-Low, High-Low, Low-High, or Not Significant using p <= 0.05.

> **Method changes.** Any change to neighbour definition, standardisation, significance threshold, or missing-value handling is an analytical-method change and requires updated tests and documentation.

6. UI and output contracts

- There is no standalone location selector. Analysis includes all units at the selected level.
- Single mode uses one week; range mode uses the inclusive slider endpoints.
- The Global Moran banner precedes both maps.
- Prevalence and LISA legends are always added by the map functions.
- The Results and Prepared Data tables support one selected-column filter and display-decimal settings.

CSV downloads should use unformatted data. Optional map PNG export depends on the installed screenshot/browser packages; the UI must fail gracefully when they are unavailable.

7. Testing and validation

The current tests cover API-POP import, configuration/pre-flight, geospatial preparation, and spatial/output helpers.

```
testthat::test_dir("tests/testthat")
```

- Test at least one country with each configured level.
- Verify leading-zero IDs and label fallback joins.
- Test single-week and multi-week population behavior.
- Test fewer than four units and zero-variance data.
- Inspect legends, popups, Global Moran banner, tables, and downloads.

8. Deployment and security

- Package raw_data GeoPackages with the deployment or mount them at the expected path.
- Preserve case-sensitive layer and column names.
- Use an approved server or desktop wrapper for restricted data.
- Test offline basemap behavior when is_offline is true.
- Avoid exposing internal file-system paths in user-facing errors on public deployments.

For shinyapps.io or container deployment, verify that GDAL/PROJ system libraries and R spatial packages are compatible with the GeoPackage files.

9. Maintenance checklist

- Run pre-flight against every configured country.
- Run automated tests.
- Start from a clean R session.
- Validate every map level and time mode.
- Confirm manuals describe GeoPackages, not obsolete RDS files.

Symptom	Likely area
Missing GeoPackage	raw_data path and Country.file_name.
Layer cannot be inspected	layer_name or GeoPackage integrity.
Join returns zero rows	csv_column, spatial_id, name_col/label_col, code_length.
Moran/LISA unavailable	Unit count, variance, missing values, or neighbour graph.
Wrong population	Deduplication source geography and repeated denominators.
Map display issue	functions_plot.R, leaflet dependencies, or offline settings.